

1 Course outline

- signal analysis
 - types of discrete-time signals, elementary properties and operations
 - Fourier transforms for discrete-time signals
- signal processing
 - linear systems
 - filters and filter design
- from DSP to machine learning
 - random signals, noise, quantization
 - adaptive filters
- sampling and interpolation
 - from analog to digital and vice versa

2 Useful sequences

Unit impulse | Delta

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}, \quad \delta_m[n] = \delta[n - m]$$

Unit step

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Exponential decay

$$x[n] = a^n u[n], \quad 0 < a < 1$$

Sinusoidal oscillation

$$x[n] = A \cos(\omega n + \phi), \quad A \in \mathbb{R}, \omega \in \mathbb{R}, \phi \in \mathbb{R}$$

Complex exponential

$$x[n] = A e^{j(\omega n + \phi)}, \quad A \in \mathbb{R} \text{ amplitude, } \phi \text{ phase, } \omega \in \mathbb{R} \text{ frequency (rad)}.$$

Key differences

- $\delta[n]$ is the most concentrated signal in time, $u[n]$ is the most concentrated in frequency.
- $\delta[n]$ has all frequencies equally present, $u[n]$ has only DC.
- $\delta[n]$ is the identity for convolution, $u[n]$ is the identity for multiplication.

Energy and Power

- $E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$
- $P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$.
- $E_x < \infty$ and $P_x = 0 \Rightarrow$ Energy signal
- $P_x < \infty$ and $E_x = \infty \Rightarrow$ Power signal

3 Basic operations

Amplitude scaling

$$y[n] = \alpha x[n]$$

Sum of signals

$$y[n] = x_1[n] + x_2[n]$$

Product of signals

$$y[n] = x_1[n] \cdot x_2[n]$$

Time shift by $k \in \mathbb{Z}$

$$y[n] = \mathcal{S}^k x[n] = x[n + k]$$

Time reversal

$$y[n] = \mathcal{R}x[n] = \mathcal{S}^{-n} x[0] = x[-n]$$

Sum of a sequence

$$\sum_{k=m}^n z^k = \begin{cases} n - m + 1 & z = 1 \\ \frac{z^m - z^{n+1}}{1 - z} & z \neq 1 \end{cases}$$

4 Vector spaces of signals

Vector space of signals

- $\ell_2(\mathbb{Z})$: space of square-summable, infinite sequences:
$$x \in \ell_2(\mathbb{Z}) \iff \|x\|^2 = \sum_{n \in \mathbb{Z}} |x[n]|^2 < \infty$$
- $L_2([a, b])$: space of square-integrable functions over the interval $[a, b]$:
$$x \in L_2([a, b]) \iff \int_a^b |x(t)|^2 dt < \infty$$

In both cases, we require vectors to have finite energy

Inner product space

- $\|x\| = \sqrt{\langle x, x \rangle}$
- $\langle x, y \rangle = \langle y, x \rangle^*$
- $\langle x, \alpha y \rangle = \alpha \langle x, y \rangle$
- $\langle \alpha x, y \rangle = \alpha^* \langle x, y \rangle$

Orthogonal basis coordinates

For orthog. basis $B = \{\mathbf{v}_k\}_{k=0}^{N-1}$ with inner product $\langle \cdot, \cdot \rangle$, the coord. of $\mathbf{x}$ in B are:

$$[\mathbf{x}]_B[k] = \frac{\langle \mathbf{v}_k, \mathbf{x} \rangle}{\langle \mathbf{v}_k, \mathbf{v}_k \rangle}, \quad \text{i.e. } \mathbf{x} = \sum_k [\mathbf{x}]_B[k] \mathbf{v}_k$$

DFT: basis $\{\mathbf{w}_k\}$, $\|\mathbf{w}_k\|^2 = N$, so $X[k] = \langle \mathbf{w}_k, \mathbf{x} \rangle$ (the N factor is absorbed into the synthesis $1/N$).

5 The Complex Exponential

Discrete-time complex exponential

$$w[n] = A e^{j(\omega n + \phi)}, \quad A \in \mathbb{R} \text{ amplitude, } \phi \text{ phase, } \omega \in \mathbb{R} \text{ frequency (rad)}.$$

Recursive form

$$w[n + 1] = w[n] e^{j\omega}: \text{ each step rotates by } \omega \text{ on the unit circle.}$$

Periodicity
 $\mathbf{w}$ is periodic iff $\exists N \in \mathbb{Z}^+, M \in \mathbb{Z}$ such that $\omega N = 2\pi M$:

$$\omega = \frac{M}{N} 2\pi, \quad M, N \in \mathbb{Z}$$

Only oscillations with **rational multiple of 2π** are periodic.

Aliasing

Key identity

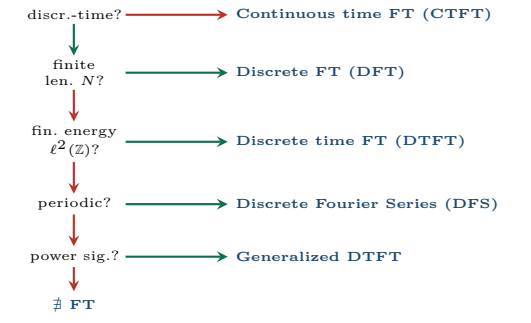
$$e^{j\theta} = e^{j(\theta + 2k\pi)} \quad \forall k \in \mathbb{Z}. \text{ Hence:}$$
$$e^{j(\omega + 2k\pi)n} = e^{j\omega n} \quad \forall k \in \mathbb{Z}.$$

Adding multiples of 2π to ω gives the **same samples**.

- $0 \leq \omega < \pi$: counterclockwise, increasing speed
- $\omega = \pi$: max speed, $+1/-1$ alternating
- $\pi < \omega < 2\pi$: more “economical” as clockwise $\omega' = \omega - 2\pi$

- Reference interval:** $[-\pi, \pi]$.
- Why complex oscillations?
- **Simpler math:** $\cos(\alpha + \beta) = \text{Re}\{e^{j\alpha} e^{j\beta}\}$
 - **Compact notation:** polar vs. two real coordinates
 - **Digital signals:** can natively be complex-valued

6 Signal analysis



7 The DFT

Fourier Basis for $\mathbb{C}^N$

Fourier basis vectors

$w_k[n] = e^{j\frac{2\pi}{N}kn}$, $n, k = 0, \dots, N-1$.
Harmonically related: $\omega_k = \frac{2\pi}{N}k$.
Note: $\mathbf{w}_k = \mathbf{w}_{N-k}^*$.

Orthogonality

$\langle \mathbf{w}_h, \mathbf{w}_k \rangle = N \delta[h-k]$, where

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n=0}^{N-1} x^*[n] y[n].$$

Basis is orthogonal but **not** orthonormal (norms = $\sqrt{N}$).

Analysis & Synthesis

DFT (analysis)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk}, \quad k = 0, \dots, N-1$$

Inverse DFT (synthesis)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}nk}, \quad n = 0, \dots, N-1$$

Elementary DFT Pairs

Impulse & constant (duality)

$\delta \xleftrightarrow{\text{DFT}} \mathbf{1}$ (most concentrated in time $\Rightarrow$ flat spectrum)
 $\mathbf{1} \xleftrightarrow{\text{DFT}} N\delta$ (most spread in time $\Rightarrow$ single freq.)

Shifted impulse

$$\delta[n-m] \xleftrightarrow{\text{DFT}} e^{-j\frac{2\pi}{N}mk} = \mathbf{w}_m^*[k]$$

Harmonic sinusoid

$\mathbf{w}_m \xleftrightarrow{\text{DFT}} N\delta_m$, where $w_m[n] = e^{j\frac{2\pi}{N}mn}$
Spread in time $\Rightarrow$ single nonzero freq. coefficient.

Rectangular (M -length rect.)

$x[n] = 1$ for $0 \leq n < M$, else 0:

$$X[k] = \frac{\sin(\pi Mk/N)}{\sin(\pi k/N)} e^{j\frac{\pi}{N}(M-1)k}$$

Zeros at $k = mN/M$ ($m \neq 0$). $M = N$: gives $N\delta$.

Time-frequency concentration duality

Most concentrated in time $\Leftrightarrow$ most spread in frequency (and vice versa). Generalizes $\delta \leftrightarrow \mathbf{1}$ and $\mathbf{1} \leftrightarrow N\delta$.

Roots of unity

The N solutions of $z^N = 1$: $z_k = e^{j\frac{2\pi}{N}k}$, $k = 0, \dots, N-1$. These are the frequencies $\omega_k = \frac{2\pi}{N}k$ satisfying $e^{j\omega_k N} = 1$.

Matrix Form

DFT matrix

$\mathbf{X} = \mathbf{W}\mathbf{x}$, $\mathbf{x} = \frac{1}{N}\mathbf{W}^H\mathbf{X}$, where $W_N = e^{-j\frac{2\pi}{N}}$:

$$\mathbf{W} = \begin{bmatrix} W_N^0 & W_N^0 & \dots & W_N^0 \\ W_N^0 & W_N^1 & \dots & W_N^{N-1} \\ \vdots & & \ddots & \vdots \\ W_N^0 & W_N^{N-1} & \dots & W_N^{(N-1)^2} \end{bmatrix}$$

$$\mathbf{W}^H \mathbf{W} = N\mathbf{I}.$$

Powers modulo N

$W_N^n = W_N^{n \bmod N}$ (aliasing property). First row and column are all 1 since $W_N^0 = 1$.

Small DFT matrices

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\mathbf{W}_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3 & W_3^2 \\ 1 & W_3^2 & W_3 \end{bmatrix}, \quad W_3 = \frac{-1-j\sqrt{3}}{2}$$

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

8 Properties

DFT Properties

Linearity

$$\text{DFT}\{\mathbf{a}\mathbf{x} + \mathbf{b}\mathbf{y}\} = a \text{DFT}\{\mathbf{x}\} + b \text{DFT}\{\mathbf{y}\}, \quad a, b \in \mathbb{C}$$

Signal decomposition (computation trick)

Decompose $\mathbf{x} = \sum_i a_i \mathbf{s}_i$ into signals with known DFTs, then use linearity.
Ex.: $\mathbf{x} = a\delta + b \cdot \mathbf{1} \Rightarrow \mathbf{X} = a \cdot \mathbf{1} + bN\delta$

Circular time shift

$$x[(n-n_0) \bmod N] \xleftrightarrow{\text{DFT}} e^{-j\frac{2\pi}{N}n_0k} X[k]$$

Frequency shift

$$x[n] e^{j\frac{2\pi}{N}k_0n} \xleftrightarrow{\text{DFT}} X[k-k_0]$$

Modulation by cosine

$$x[n] \cos\left(\frac{2\pi}{N}k_0n\right) \xleftrightarrow{\text{DFT}} \frac{X[k-k_0] + X[k+k_0]}{2}$$

Time/frequency reversal

$$\begin{aligned} \text{DFT}\{\mathcal{R}\mathbf{x}\} &= \mathcal{R}\mathbf{X} \\ \text{DFT}\{\mathbf{x}^*\} &= \mathcal{R}\mathbf{X}^* \end{aligned}$$

Parseval's theorem (DFT)

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 = \|\mathbf{x}\|_2^2$$

Energy is preserved across the DFT change of basis ($1/N$ from non-normalized basis). $|X[k]|^2/N$ = energy at frequency ω_k . Useful: $\|\mathbf{x} - \mathbf{y}\|_2^2 = \frac{1}{N}\|\mathbf{X} - \mathbf{Y}\|_2^2$.

Symmetries of the DFT

A signal $\mathbf{x}$ is **symmetric** if $x[n] = x[N-n]$.
A signal is **Hermitian-symmetric** if $x[n] = x^*[N-n]$.

Symmetry summary

Time domain $x[n]$	Freq. domain $X[k]$
Real	$X[-k] = X^*[k]$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

9 Reading & Plotting the DFT

DFT magnitude plot

Index k encodes frequency $\omega_k = \frac{2\pi}{N}k$:

- $k = 0$: DC (zero frequency)
- $k = 1 \dots \lfloor N/2 \rfloor$: positive (CCW) frequencies, increasing
- $k = \lfloor N/2 \rfloor$: maximum $\omega = \pi$ (fastest oscillation)
- $k = \lfloor N/2 \rfloor + 1 \dots N-1$: negative (CW) frequencies, decreasing

For real $\mathbf{x}$: often show only $k = 0 \dots N/2$ (symmetric).

DFT phase

Phase is 2π -periodic, plotted over $[-\pi, \pi]$. Values outside are **wrapped**. Phase offset θ in time shifts all components; same magnitude DFT can give different time-domain shapes.

Frequency axis in Hz

If signal sampled at $F_s = N/T$ Hz (T = duration in seconds):

$$k \rightarrow \frac{F_s}{N}k \text{ Hz}, \quad 0 \leq k \leq N/2$$

$$f_{\max} = \frac{F_s}{2} \text{ (Nyquist)}$$

For $k > N/2$: $k \rightarrow \frac{F_s}{2} - \frac{F_s}{N}k$ Hz (negative freqs).

10 DFS & DFT Periodicity

DFT implies N-periodicity

The inverse DFT formula is valid for **all** $n \in \mathbb{Z}$:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}nk} = x[n \bmod N]$$

Both x and X are implicitly N -periodic.

DFS (Discrete Fourier Series)

For an N -periodic signal $\tilde{x}$:

$$\tilde{\mathbf{X}} = \text{DFS}\{\tilde{\mathbf{x}}\} = \text{DFT}\{\tilde{x}[0], \dots, \tilde{x}[N-1]\}$$

DFS $\equiv$ DFT: same formulas, different interpretation (explicit periodicity).

Circular shift (DFS)

$$\tilde{x}[n-n_0] \xleftrightarrow{\text{DFS}} e^{-j\frac{2\pi}{N}n_0k} \tilde{X}[k]$$

Time shifts are circular
With finite-length signals, time shifts wrap around modulo N . A linear phase multiplication on DFT coefficients computes the circular shift of the signal.

11 The DTFT

DTFT (analysis)
For infinite-length sequence $\mathbf{x}$:
$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$
$X(\omega)$ is a continuous, 2π-periodic function of $\omega \in \mathbb{R}$.

Inverse DTFT (synthesis)
$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$
Uses $\int_{-\pi}^{\pi} e^{j\omega(n-k)} d\omega = 2\pi \delta[n-k]$.

Existence & Convergence

Existence: absolutely summable $\Rightarrow$ DTFT
If $\sum_{n=-\infty}^{\infty} x[n] < \infty$, then $X(\omega)$ exists, is continuous, and $ X_N(\omega) \rightarrow X(\omega)$ uniformly.

Square-summable (finite energy)
If $\mathbf{x} \in \ell_2(\mathbb{Z})$: $\sum x[n] ^2 < \infty$, DTFT exists in mean-square sense:
$\lim_{N \rightarrow \infty} \int_{-\pi}^{\pi} X_N(\omega) - X(\omega) ^2 d\omega = 0$
$X(\omega)$ may not be continuous (Gibbs phenomenon possible).

DTFT as Limit of DFT
As $N \rightarrow \infty$: $\omega_N = 2\pi/N \rightarrow 0$, DFT samples become dense over $[0, 2\pi]$. By substituting $\omega = 2\pi k/N$:

$$X[k] \rightarrow X\left(\frac{2\pi k}{N}\right)$$

Also: $\tilde{X}_N[k] = X\left(\frac{2\pi}{N}k\right)$ (DFS as samples of DTFT).

12 DTFT Pairs & Properties
Elementary DTFT Pairs
Impulse
DTFT $\{\delta_m\} = e^{-j\omega m}$ Most concentrated in time $\Rightarrow$ spread uniformly in freq.

Rectangular signal
$x[n] = \begin{cases} 1 & 0 \leq n < M \\ 0 & \text{otherwise} \end{cases}$
$X(\omega) = \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j\omega(M-1)/2}$
Zeros at $\omega = 2\pi m/M$, $m \neq 0$.

Exponential decay ($ a < 1$)
$x[n] = a^n u[n]$ ($u[n]$: unit step)
$X(\omega) = \frac{1}{1 - a e^{-j\omega}}$
$ X(\omega) ^2 = \frac{1}{1 + a ^2 - 2a \cos \omega}$

DTFT Properties
Linearity
DTFT $\{ax + by\} = a\mathbf{X} + b\mathbf{Y}$, $a, b \in \mathbb{C}$
Time shift
$x[n - n_0] \xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(\omega)$

Frequency shift (modulation)
$x[n] e^{j\omega_0 n} \xleftrightarrow{\text{DTFT}} X(\omega - \omega_0)$
$x[n] \cos(\omega_0 n) \xleftrightarrow{\text{DTFT}} \tfrac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$

Time & frequency reversal
DTFT $\{\mathcal{R}\mathbf{x}\} = \mathcal{R}\mathbf{X}$ DTFT $\{\mathbf{x}^*\} = \mathcal{R}\mathbf{X}^*$

Symmetries of the DTFT
A signal is **even** if $x[n] = x[-n]$, **odd** if $x[n] = -x[-n]$.
A signal is **Hermitian-symmetric** if $x[n] = x^*[-n]$.

Symmetry summary	
Time domain $x[n]$	Freq. domain $X(\omega)$
Real	$X(-\omega) = X^*(\omega)$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

Parseval's theorem (DTFT)
$\sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) ^2 d\omega$

Reading DTFT plots
DTFT frequency axis
Plotted over $[-\pi, \pi]$: $\omega = 0$: DC $\omega \in (0, \pi]$: positive (CCW) = high freqs near $\pm\pi$ $\omega \in [-\pi, 0)$: negative (CW) = high freqs near $\pm\pi$ - Low freqs near 0, high freqs near $\pm\pi$ DTFT is 2π -periodic; plots outside $[-\pi, \pi]$ repeat.

Generalized DTFT (Power Signals)
Via functionals for infinite-energy signals:

Constant & complex exponential
DTFT $\{1\} = \delta(\omega)$ (Dirac delta at $\omega = 0$) $e^{j\omega_0 n} \xleftrightarrow{\text{DTFT}} \delta(\omega - \omega_0)$

Sinusoids
$\cos(\omega_0 n) \xleftrightarrow{\text{DTFT}} \tfrac{1}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$ $\sin(\omega_0 n) \xleftrightarrow{\text{DTFT}} \tfrac{-j}{2} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$

Unit step
$U(\omega) = \frac{1}{1 - e^{-j\omega}} + \tfrac{1}{2} \delta(\omega)$

13 DFT vs. DTFT vs. DFS		
	Input	Output
DFT	finite, length N	vector $\in \mathbb{C}^N$
DFS	N -periodic	N -periodic
DTFT	infinite (ℓ_2)	2π -periodic function

DFT/DFS equivalence
DFS of an N -periodic signal = DFT of one period. Both use identical formulas (4.10) and (4.11); only the interpretation of periodicity differs.

DTFT as DFS limit
$\tilde{x}_N[n] = \sum_{p=-\infty}^{\infty} x[n + pN]$ (periodization). $\tilde{X}_N[k] = X\left(\frac{2\pi}{N}k\right)$: DFS samples the DTFT. As $N \rightarrow \infty$: $\tilde{x}_N \rightarrow \mathbf{x}$ and samples fill $[0, 2\pi]$.

14 The FFT
Complexity comparison
- Naive DFT: $O(N^2)$ multiplications - FFT: $O(N \log N)$ operations - For $N = 10,000$: $\sim 40,000$ vs. > 100 million ops

Decimation in time (Cooley-Tukey)
Split $N = 2^L$ input into even/odd: $x_e[n] = x[2n]$, $x_o[n] = x[2n + 1]$, $n = 0, \dots, \frac{N}{2} - 1$ Output halves: $X_a[k] = X[k]$ and $X_b[k] = X[k + \frac{N}{2}]$, $k = 0, \dots, \frac{N}{2} - 1$ Butterfly equations: $X_a[k] = X_e[k] + e^{j\frac{2\pi}{N}k} X_o[k]$ $X_b[k] = X_e[k] - e^{j\frac{2\pi}{N}k} X_o[k]$ where X_e, X_o are the $(N/2)$ -point DFTs of x_e, x_o .
Cost breakdown
$2(N/2)^2 = N^2/2$ mults for $\mathbf{X}_e$ and $\mathbf{X}_o$ $N/2$ mults by twiddle factors for $\mathbf{X}_o$ - Second half is free (just sign change) Recursion gives $O(N \log_2 N)$ total.

15 Spectrograms (STFT)
Short-Time Fourier Transform
Analyse a signal $\mathbf{x}$ of length L time-locally: - Slide a window of size W to get $\lfloor L/W \rfloor$ non-overlapping bins - Compute a W-point DFT in each bin - Result: 2D time $\times$ frequency representation

Kronecker delta in spectrogram
DFT of a length- W Kronecker delta $\delta[n - m]$ (even shifted): $ X[k] = 1$ for all k . $\Rightarrow$ In a spectrogram with window W : the bin containing m has constant magnitude W across all frequencies; all other bins are 0 .

Window size trade-off
Large W : good frequency resolution, poor time resolution. Small W : good time resolution, poor frequency resolution. This is an instance of the time–frequency concentration duality.

16 Discrete-Time Filters

LTI Systems

Linearity & time-invariance

Let $y[n] = \mathcal{H}\{x[n]\}$.
Linearity ($\forall \alpha, \beta \in \mathbb{C}, \forall x_1, x_2$):

$$\mathcal{H}\{\alpha x_1[n] + \beta x_2[n]\} \stackrel{?}{=} \alpha \mathcal{H}\{x_1[n]\} + \beta \mathcal{H}\{x_2[n]\}.$$

Time-Invariance ($\forall n_0 \in \mathbb{Z}, \forall x$):

$$\underbrace{\mathcal{H}\{x[n - n_0]\}}_{x[\cdot] \leftarrow x[\cdot - n_0]} \stackrel{?}{=} \underbrace{y[n - n_0]}_{n \leftarrow n - n_0}.$$

Moving Average (MA)

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n - k]$$

Causal FIR; $h[n] = \frac{1}{M}$ for $0 \leq n < M$, else 0.
Requires $M - 1$ additions and 1 multiplication per output sample.

Zero-centered (noncausal) MA

$$y[n] = \frac{1}{2L + 1} \sum_{k=-L}^L x[n - k]$$

Noncausal; eliminates the processing lag of the causal version. Causal implementation: delay output by L samples. Causal up to a delay $\Rightarrow$ always FIR, always BIBO stable.

Leaky Integrator (LI)

$$y[n] = \lambda y[n - 1] + (1 - \lambda)x[n], \quad |\lambda| < 1$$

Causal IIR; $h[n] = (1 - \lambda)\lambda^n u[n]$.
Approximates MA of length M with $\lambda = \frac{M-1}{M}$.
Requires 2 multiplications and 1 addition per sample (constant cost).

LI BIBO stability

LI is BIBO stable $\iff |\lambda| < 1$.

Proof: $\sum_{n=0}^{\infty} |h[n]| = |1 - \lambda| \sum_{n=0}^{\infty} |\lambda|^n = \frac{|1 - \lambda|}{1 - |\lambda|} < \infty$
iff $|\lambda| < 1$.
As $\lambda \rightarrow 1$: smoothing power $\uparrow$ but output diverges at $\lambda = 1$.

Impulse Response & Convolution

Impulse response

$\mathbf{h} = \mathcal{H}\delta$. An LTI system is *fully* described by $\mathbf{h}$.

Convolution $y = x * h$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Well-defined when $\sum_k x[k]y[n - k]$ is finite $\forall n$ (e.g., both abs. summable, or one abs. summable and one bounded).

Convolution properties

Linearity: $\mathbf{x} * (\alpha \mathbf{y} + \beta \mathbf{w}) = \alpha(\mathbf{x} * \mathbf{y}) + \beta(\mathbf{x} * \mathbf{w})$
Time-shift: $\mathbf{x} * (\mathcal{S}^k \mathbf{y}) = \mathcal{S}^k(\mathbf{x} * \mathbf{y}) = (\mathcal{S}^k \mathbf{x}) * \mathbf{y}$
Commutativity: $\mathbf{x} * \mathbf{h} = \mathbf{h} * \mathbf{x}$
Associativity: $(\mathbf{x} * \mathbf{h}) * \mathbf{w} = \mathbf{x} * (\mathbf{h} * \mathbf{w})$ (abs. summable)
Cascade: two stable filters in series $\equiv$ one filter with $\mathbf{h} = \mathbf{h}_1 * \mathbf{h}_2$; order is irrelevant.

Computing $y[n]$ algorithmically

1. Time-reverse h : $h[k] \rightarrow h[-k]$
2. Shift by n : $h[-k] \rightarrow h[n - k]$
3. Multiply element-wise with $x[k]$ and sum.

Filter Classification (Time Domain)

FIR vs IIR

- **FIR:** finite-support $\mathbf{h}$; no feedback; *always* BIBO stable.
- **IIR:** infinite-support $\mathbf{h}$; realizable IIRs have one-sided $\mathbf{h}$ via feedback paths.

Causality

$\mathcal{H}$ is (**strictly**) **causal** iff $h[n] = 0$ for $n < n_0$, some $n_0 \geq 0$. Output depends only on past/present inputs. When $\mathbf{h}$ is not strictly causal:

1. **Causal up to a delay:** $h[n] = 0$ for $n < n_0$, $n_0 < 0$ (FIR is always this).
2. **Anticausal:** $h[n] = 0$ for $n > 0$; offline applications only.
3. **Ideal:** infinite two-sided support — *not realizable* in practice; used as theoretical benchmark.

BIBO Stability

Bounded Input $\Rightarrow$ Bounded Output. LTI is BIBO stable $\iff \mathbf{h}$ abs. summable:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

($\Leftarrow$) **Suff.:** Let $|x[n]| \leq L$. By triangle inequality then bounded input:

$$|y[n]| \leq \sum_k |h[k]| \underbrace{|x[n - k]|}_{\leq L} \leq L \sum_k |h[k]| < \infty$$

($\Rightarrow$) **Nec.:** If $\sum |h[k]| = \infty$, take bounded $x[n] = \text{sgn}(h[-n])$; then $y[0] = \sum_k |h[k]| = \infty$.

FIR always BIBO stable

FIR impulse response has finite support $\Rightarrow \sum |h[k]| < \infty$ always holds $\Rightarrow$ FIR filters are **unconditionally BIBO stable**.

Frequency Response

Eigensequence property

For $w[n] = e^{j\omega_0 n}$: $\mathcal{H}\mathbf{w} = H(\omega_0)\mathbf{w}$, where

$$H(\omega) = \text{DTFT}\{h\} = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$$

$H(\omega_0) = A_0 e^{j\theta_0}$: $A_0 = |H(\omega_0)|$ scales amplitude, $\theta_0 = \angle H(\omega_0)$ shifts phase.
LTI cannot create new frequency content.

Convolution Theorem

$$y = x * h \iff Y(\omega) = X(\omega) \cdot H(\omega)$$

Filtering = pointwise multiplication in frequency domain.

Filter Categories

- **Lowpass:** $|H(\omega)|$ large near $\omega = 0$, small near $\pm\pi$
- **Highpass:** $|H(\omega)|$ large near $\omega = \pm\pi$, small near 0
- **Bandpass:** $|H(\omega)|$ large near $\omega = \pm\omega_c$
- **Allpass:** $|H(\omega)| = \text{const}$ (phase only)

Passband: where $|H| \approx 1$; **Stopband:** where $|H| \approx 0$.

Phase Response & Group Delay

Linear phase

$\angle H(\omega) = -\omega d$, $d \in \mathbb{R}$: uniform delay d samples for all frequencies. All realizable linear-phase filters are FIR.
Generalized: $H(\omega) = A(\omega)e^{-j(\omega d + \phi)}$, $A(\omega) \in \mathbb{R}$.

Group delay

$$\text{grd}\{H(\omega)\} = -\frac{d \angle H(\omega)}{d\omega}$$

Constant for linear-phase filters. For a group of freqs around ω_0 , the group delay gives the time delay.

Key Filter Frequency Responses

Moving Average $H(\omega)$

$$H(\omega) = \frac{1}{M} \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j \frac{M-1}{2} \omega}$$

Lowpass; $M - 1$ zero crossings at $\omega = 2\pi m/M$ ($m \neq 0$). Generalized linear phase with delay $(M - 1)/2$.

Leaky Integrator $H(\omega)$

$$H(\omega) = \frac{1 - \lambda}{1 - \lambda e^{-j\omega}}$$
$$|H(\omega)|^2 = \frac{(1 - \lambda)^2}{1 + \lambda^2 - 2\lambda \cos \omega}$$
$$\angle H(\omega) = \arctan \left[\frac{-\lambda \sin \omega}{1 - \lambda \cos \omega} \right]$$
$$\text{grd}\{H(\omega)\} = \frac{\lambda \cos \omega - \lambda^2}{1 + \lambda^2 - 2\lambda \cos \omega}$$

Lowpass; bandwidth decreases as $\lambda \rightarrow 1$.

Ideal Filters

sinc & rect

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}, \quad \text{sinc}(0) = 1. \quad \text{rect}(x) = \begin{cases} 1 & |x| \leq \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$$

Fundamental sinc/rect DTFT pair

$$\frac{\omega_p}{\pi} \text{sinc}\left(\frac{\omega_p}{\pi} n\right) \stackrel{\text{DTFT}}{\longleftrightarrow} \text{rect}\left(\frac{\omega}{2\omega_p}\right)$$

As $\omega_p \rightarrow \pi$: recovers $\delta \leftrightarrow \mathbf{1}$. As $\omega_p \rightarrow 0$: $\mathbf{1} \leftrightarrow \tilde{\delta}(\omega)$.

Ideal Lowpass (cutoff ω_p)

$$H_L(\omega) = \begin{cases} 1 & |\omega| \leq \omega_p \\ 0 & \text{else} \end{cases}, \quad h_L[n] = \frac{\omega_p}{\pi} \text{sinc}\left(\frac{\omega_p}{\pi} n\right)$$

Infinite two-sided support $\Rightarrow$ **not realizable**.

Ideal Highpass & Bandpass

$$H_H(\omega) = 1 - H_L(\omega), \quad h_H[n] = \delta[n] - h_L[n]$$

$$h_B[n] = 2 \cos(\omega_c n) \frac{\omega_p}{\pi} \operatorname{sinc}\left(\frac{\omega_p}{\pi} n\right) \text{ (BW } 2\omega_p, \text{ center } \omega_c)$$

Hilbert Filter (allpass)

$$H_F(\omega) = \begin{cases} -j & 0 \leq \omega < \pi \\ +j & -\pi \leq \omega < 0 \end{cases}, \quad |H_F| = 1$$

$$h_F[n] = \begin{cases} 2/(n\pi) & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

Realizable Filters: CCDE

Constant-Coefficient Difference Equation

$$\sum_{k=0}^{N-1} a_k y[n-k] = \sum_{k=0}^{M-1} b_k x[n-k], \quad a_0 = 1$$

$$y[n] = \sum_{k=0}^{M-1} b_k x[n-k] - \sum_{k=1}^{N-1} a_k y[n-k]$$

FIR: $a_k = 0$ for $k \geq 1$. **IIR:** some $a_k \neq 0$ ($k \geq 1$, feedback).

Zero initial conditions

For a recursive filter to be uniquely defined and LTI, $\exists n_0$ such that:

- $x[n] = 0$ for all $n < n_0$ (input off before start)
- $y[n] = 0$ for all $n < n_0$ (output zero before start)
- all memory elements = 0 at $n = n_0$

Nonzero initial memory breaks uniqueness and time-invariance.

17 Quick Reference

Geometric sum

$$\sum_{n=0}^{N-1} r^n = \frac{1 - r^N}{1 - r}, \quad r \neq 1$$

Used in rectangular DFT derivation; also the orthogonality of Fourier vectors (set $r = e^{j \frac{2\pi}{N} (k-h)}$, $k \neq h$).

Euler & trig

- $(e^{j\theta})^* = e^{-j\theta}$
- $\cos(a + b) = \cos a \cos b - \sin a \sin b$
- $\sin(a + b) = \sin a \cos b + \cos a \sin b$
- $\sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)]$
- $\cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$
- $\sin a \cos b = \frac{1}{2} [\sin(a + b) + \sin(a - b)]$

Complex number tricks

- $z_1 = z_2^* \Rightarrow |z_1| = |z_2|$ ($|z| = |z^*|$ always)
- $z + z^* = 2 \operatorname{Re}\{z\}$, $z - z^* = 2j \operatorname{Im}\{z\}$
- $z = z^* \iff z \in \mathbb{R} \iff \operatorname{Im}\{z\} = 0$
- $|z_1 z_2| = |z_1| |z_2|$, $\angle(z_1 z_2) = \angle z_1 + \angle z_2$
- $|z|^2 = z \cdot z^*$

Inner product in $\mathbb{C}^N$

Coefficients: $X[k] = \langle \mathbf{w}_k, \mathbf{x} \rangle$ (projection onto basis).

DTFT inner product in $L_2([-\pi, \pi])$

$$\langle \mathbf{F}, \mathbf{G} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} F^*(\omega) G(\omega) d\omega$$

Set $\{e^{-j\omega n}\}_{n \in \mathbb{Z}}$ forms an orthogonal basis.

$$x[n] = \langle e^{j\omega n}, X(\omega) \rangle$$

Key warnings

DFT normalization

Analysis: no $1/N$ factor. **Synthesis:** $1/N$ factor.
Convention: $\mathbf{W}^H \mathbf{W} = N \mathbf{I}$.

Phase wrapping

Phase plots show $\angle X[k] \in [-\pi, \pi]$ (wrapped). Apparent discontinuities are $\pm\pi$ jumps, not real.

Real signal DFT

For real $\mathbf{x}$: $X[k] = X^*[N - k]$. Only first $\lfloor N/2 \rfloor + 1$ coefficients are independent. Magnitude symmetric around $k = N/2$.

Circular vs. linear shift

DFT/DFS time shifts are always **circular** (modulo N). For aperiodic finite signals, multiplying by phase in freq domain gives circular shift.